

block.

6

$$P = h\rho g$$

$$P_R = x\rho_R g = 4 \dots (1)$$

$$P_Q = h2\rho_R g = 9 \dots (2)$$

$$\frac{(2)}{(1)} = \frac{2h}{x} = \frac{9}{4}$$

$$h = \frac{9}{8}x$$

7

$$W_{balance} - T = m_{balance}a$$